

1. Administrative & Study Identification

Full Study Title: A Phase 2b, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Tulisokibart in Participants With Rheumatoid Arthritis **Short Title/Acronym:** MK-7240-014 **Protocol Number:** MK-7240-014 **NCT Number:** NCT07176390 **Trial Status:** Recruiting **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Tulisokibart (MK-7240), an investigational humanized monoclonal antibody directed to tumor necrosis factor (TNF)-like cytokine 1A (TL1A) **Phase of Development:** Phase 2 (Phase 2b) **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if at least one dose of the investigational medicine tulisokibart works better than a placebo to lessen rheumatoid arthritis (RA) symptoms in participants taking methotrexate (MTX). **Secondary Objectives:** To evaluate the efficacy based on further clinical response criteria and to evaluate the long-term safety, tolerability, and sustained efficacy of tulisokibart during the 116-week Long-term Extension (LTE).

2.2 Background & Rationale Methotrexate (MTX; trade names include Rheumatrex, Emtexate, Trexall, Maxtrex, Jylamvo, Reditrex, Xatmep, Otrexup, Rasuvo) is a standard treatment for RA. However, MTX and other current treatments may not work well to treat RA symptoms for many people. Tulisokibart is an investigational monoclonal antibody targeting TL1A, a cytokine that is an important mediator of inflammation. This study aims to find out if tulisokibart can effectively treat symptoms of active RA in people who have an inadequate response to MTX or up to two classes of biologic disease-modifying antirheumatic drugs (bDMARDs).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target **SN\$:** 182 participants **Number of Sites:** Global multicenter **Estimated Study Start:** 08-Oct-2025 **Estimated Primary Completion:** 08-Mar-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 80 years. 2. Has a clinical diagnosis of rheumatoid arthritis (RA) and fulfillment of 2010 American College of Rheumatology (ACR)/European League Against Rheumatism (EULAR) classification criteria. 3. Has active disease defined as ≥ 6 tender joints (based on 68 joints) and ≥ 6 swollen joints (based on 66 joints). 4. Has current treatment with oral or parenteral methotrexate (MTX) therapy. 5. Has a history of one of the following: a) biologic disease-modifying antirheumatic drug (bDMARD) naïve, or b) bDMARD-Inadequate Response (IR)/intolerant up to a maximum of 2 classes of bDMARDS.

4.2 Exclusion Criteria 1. Has any arthritis with onset before age 17 years or current diagnosis of inflammatory joint disease other than RA (such as, but not limited to, psoriatic arthritis, systemic lupus erythematosus, gout, systemic sclerosis, myositis, pseudogout). 2. Has a history of cancer (except fully treated nonmelanoma skin cancers or cervical carcinoma in situ after complete surgical removal) and is disease-free for < 5 years before randomization. 3. Has any active infection. 4. Has known allergies, hypersensitivity, or intolerance to tulisokibart or its excipients.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple different doses of tulisokibart will be evaluated against a matching placebo. **Route of Administration:** Intravenous (IV) or Subcutaneous (SC) (standard modalities for monoclonal antibodies). **Duration of Treatment:** The study consists of a 12-week Placebo-controlled Period followed by a 116-week Long-term Extension (LTE). The LTE is composed of a 44-week Main Extension and a 72-week Optional Extension.

5.2 Concomitant & Prohibited Therapy Permitted: Background MTX (oral or parenteral) must be maintained. **Prohibited:** Other biologic DMARDs outside of the investigational product are restricted during the trial assessment periods.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Informed Consent, Medical History, Baseline Joint Assessment (68 tender / 66 swollen joint count)
2	0	Randomization, First Dose of Tulisokibart/Placebo
3	Week 12	End of Placebo-controlled Period, Primary Efficacy Assessment
4	Week 128	Completion of 116-week LTE (44-week Main + 72-week Optional Extension), Final Assessments, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of Participants Achieving American College of Rheumatology 20% Response Criteria (ACR20) at Week 12. **Measurement Method:** ACR20 response is defined as a $\geq 20\%$ improvement in swollen joint count (66 joints) and tender joint count (68 joints) and $\geq 20\%$ improvement in ≥ 3 of the following 5 assessments: Health Assessment Questionnaire Disability Index (HAQ-DI), Physician Global Assessment of Disease Activity (PGA), Patient's Global Assessment of Disease Activity (PtGA), Patient assessment of Pain Severity, and High-sensitivity C-reactive protein (hsCRP) serum values.

7.2 Secondary & Exploratory Endpoints 1. Proportion of Participants Achieving American College of Rheumatology 50% Response Criteria (ACR50) at Week 12. 2. Proportion of Participants Achieving American College of Rheumatology 70% Response Criteria (ACR70) at Week 12. 3. Proportion of Participants Achieving Low Disease Activity (LDA) based on Disease Activity Score 28 Using C-Reactive Protein (DAS28-CRP). 4. Long-term safety, tolerability, and sustained symptom resolution throughout the 116-week Long-term Extension phase.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via clinical evaluations, regular physical examinations, and patient reports across the 128-week study period. **Serious Adverse Events (SAEs):** Protocol-defined timelines (typically 24 hours) for reporting severe or life-threatening adverse events to the Sponsor. **Data Monitoring Committee (DMC):** Independent safety monitoring mechanisms managed per Merck's clinical protocols.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Designed with 182 target enrollments to provide sufficient statistical power to detect a significant difference between the tulisokibart dose arms and placebo in lessening RA symptoms. **Handling of Missing Data:** Non-responder imputation (NRI) or Multiple Imputation applied for primary categorical or continuous efficacy endpoints at Week 12.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to

ensure integrity of patient selection, dosing safety, and joint assessment blinding. **Audit Trail:** Validated eCRF tracking and electronic data capture for clinical joint counts and patient-reported outcomes.

11. Study Identifiers & Governance

Primary ID: MK-7240-014 **Posting ID:** 1111142057381689332 **Registry Number:** NCT07176390 **Trial Status:** Recruiting **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC (Merck) **Study Title:** A Clinical Study of Tulsokibart (MK-7240) to Treat Rheumatoid Arthritis (RA) (MK-7240-014) **Official Regulatory Title:** A Phase 2b, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Tulsokibart in Participants With Rheumatoid Arthritis

12. Clinical Framework (Rheumatoid Arthritis Specific)

Primary Condition: Rheumatoid Arthritis (RA) **Diagnosis Threshold:** Fulfillment of 2010 ACR/EULAR classification criteria with ≥ 6 tender joints (based on 68) and ≥ 6 swollen joints (based on 66) **Medical Methodology:** Concomitant use of standard background methotrexate (MTX) paired with T1A-targeted monoclonal antibody therapy **Optimization Target:** Significant reduction in inflammatory joint symptoms and induction of clinical remission

13. Study Procedures & Methodology

Management Pathway: 12-week placebo-controlled induction transitioning to a long-term (up to 116-week) active extension. **Education Protocol (HCP):** Blinded joint assessor calibration (66/68 joint counts) and biologic safety training. **Education Protocol (Patient):** Informed consent outlining placebo usage, symptom tracking, and extension phase requirements. **Quality Audit Mechanism:** Source data verification for bDMARD failure/naïve status and MTX background therapy continuity.

14. Observation & Follow-Up Schedule

Total Duration: Up to 128 weeks (12-week placebo-controlled core period + 116-week Long Term Extension) **Onsite Visit Cadence:** Pre-defined milestones aligning with the Week 12 primary endpoint, the Week 56 (end of Main Extension) milestone, and the Week 128 (end of Optional Extension) closeout **Remote Monitoring Frequency:** Regular intervals per Merck's global clinical development monitoring standards. **Checklist Review Interval:** Routine reviews of MTX compliance and joint status assessments.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					